

Dynamic Guideline: WHO-Based Maternal Health - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most Recent Guideline Identified:

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO maternal health guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Adolescents and young women of reproductive age, pregnant and postpartum women, healthcare providers, and policymakers
- **Scope and Objectives:** This case study documents the process and outcomes of integrating meaningful youth engagement into the WHO maternal health guideline development process, ensuring that the perspectives of young people are reflected in recommendations affecting their reproductive and maternal health outcomes.

Five WHO Guidelines Synthesized:

1. **WHO recommendations on maternal and newborn care for a positive postnatal experience (2022)**
 - Comprehensive guidance on postnatal care covering the period from birth through six weeks postpartum, addressing maternal and newborn health assessments, nutrition, mental health, and health system interventions.
2. **WHO recommendations on antenatal care for a positive pregnancy experience (2016, updated 2023)**
 - A person-centred framework of 49 recommendations covering nutritional interventions, maternal and fetal assessment, preventive measures, interventions for common physiological symptoms, and health systems interventions to improve antenatal care quality and experience.
3. **WHO recommendations: intrapartum care for a positive childbirth experience (2018, updated 2023)**
 - Recommendations on care throughout labour and childbirth, including respectful maternity care, effective communication, companionship during labour, pain management, labour augmentation, and management of the second and third stages of labour.
4. **Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage (2024)**

- Updated evidence-based recommendations on uterotonics for PPH prevention, tranexamic acid use, uterine balloon tamponade, non-pneumatic anti-shock garments, and surgical interventions for PPH management.

5. Meaningful youth engagement in a WHO guideline development process: case study of WHO maternal health guideline (29 April 2026)

- Documents the methodology and impact of incorporating youth voices into WHO maternal health guideline development, with recommendations on institutionalising youth engagement in future guideline processes.

1.2 Key Recommendations

A. Postnatal Care Recommendations

Recommendation	GRADE Evidence Quality	Strength
A minimum of four postnatal contacts is recommended for all mothers and newborns, with the first contact within 24 hours of birth	Moderate	Strong
Assessment of postpartum bleeding, uterine contraction, fundal height, temperature, and blood pressure during each postnatal contact	Moderate	Strong
Screening for postpartum depression using a validated tool at each postnatal contact	Low	Conditional
Exclusive breastfeeding counselling and support for all mothers from birth through 6 months	High	Strong
Iron and folic acid supplementation for postpartum women in settings with high anaemia prevalence	Moderate	Conditional
Kangaroo mother care for all newborns initiated immediately after birth	High	Strong

B. Antenatal Care Recommendations

Recommendation	GRADE Evidence Quality	Strength
A minimum of eight antenatal care contacts is recommended to reduce perinatal mortality and improve women's experience of care	Moderate	Strong
Daily oral iron (30–60 mg elemental iron) and folic acid (400 µg) supplementation for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth	High	Strong
Calcium supplementation (1.5–2.0 g oral elemental calcium/day) in populations with low dietary calcium intake to reduce the risk of pre-eclampsia	High	Strong

One ultrasound scan before 24 weeks of gestation to estimate gestational age, detect multiple pregnancies, and identify fetal anomalies	Moderate	Conditional
Midwife-led continuity of care models recommended where well-functioning midwifery programmes exist	High	Conditional
Screening for gestational diabetes mellitus at 24–28 weeks of gestation using locally appropriate criteria	Low	Conditional

C. Intrapartum Care Recommendations

Recommendation	GRADE Evidence Quality	Strength
Respectful maternity care, including freedom from mistreatment, right to informed consent, and privacy, for all women throughout labour and childbirth	Low	Strong
Companionship of choice during labour and childbirth	Moderate	Strong
Use of partograph for routine labour monitoring	Low	Conditional
Delayed cord clamping (not earlier than 1 minute after birth) for improved maternal and neonatal outcomes	Moderate	Strong
Active management of the third stage of labour with uterotonics for all vaginal births	High	Strong
Oxytocin (10 IU, IV or IM) as the recommended uterotonic agent for prevention of PPH	High	Strong

D. Postpartum Haemorrhage Prevention and Treatment

Recommendation	GRADE Evidence Quality	Strength
Oxytocin (10 IU, IV or IM) as first-line uterotonic for PPH prevention during the third stage of labour	High	Strong
Heat-stable carbetocin (100 µg, IM) as an alternative in settings where oxytocin is unavailable or cold-chain requirements cannot be met	Moderate	Conditional
Tranexamic acid (1 g IV) administered within 3 hours of birth onset for PPH treatment, in addition to standard care	High	Strong
Uterine balloon tamponade as a temporising measure for PPH unresponsive to uterotonics	Low	Conditional
Non-pneumatic anti-shock garment (NASG) as a temporising measure for hypovolaemic shock during transfer to definitive care	Low	Conditional
Bimanual uterine compression as a first-line temporising measure for atonic PPH while awaiting definitive treatment	Very Low	Conditional

E. Youth Engagement in Guideline Development

Recommendation	GRADE Evidence Quality	Strength
Institutionalise meaningful youth engagement in all stages of maternal health guideline development, from scoping to dissemination	Low	Conditional
Provide capacity-building and mentorship for young people participating in guideline development groups	Very Low	Conditional
Ensure representation of diverse youth populations, including adolescents from low-resource settings, in guideline panels	Very Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of randomised controlled trials, observational studies, and qualitative evidence syntheses were conducted for each recommendation domain. The Cochrane Pregnancy and Childbirth Group contributed updated systematic reviews. Evidence-to-decision (EtD) frameworks were used to move from evidence to recommendations, considering benefits and harms, values and preferences, resource use, equity, acceptability, and feasibility.
- **Consensus Method:** The Guideline Development Group (GDG) comprised international experts in maternal health, midwifery, obstetrics, neonatology, public health, and patient/community representatives. Recommendations were formulated through structured consensus using the GRADE approach. For the 2026 case study, a dedicated youth advisory panel was convened alongside the GDG, with youth representatives having voting rights on recommendations directly affecting adolescent maternal health.
- **Conflict of Interest Management:** All GDG members submitted declarations of interest (DOI) reviewed by the WHO Steering Group. Members with significant conflicts were recused from relevant discussions and voting. The process followed the WHO Handbook for Guideline Development (2nd edition) and the WHO Framework for Engagement with Non-State Actors.

II. Evidence Summary After WHO Latest Guideline Release (Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Systematic search of published and preprint literature from the date of the most recent WHO consolidated PPH guideline (2024) through 29 April 2026, focusing on maternal health interventions spanning antenatal, intrapartum, and postnatal care, with particular emphasis on PPH prevention and treatment, pre-eclampsia/eclampsia management, maternal sepsis, and maternal mental health.
- **Databases Searched:**
 - PubMed/MEDLINE

- Cochrane Library (CENTRAL)
- Embase
- WHO Global Index Medicus
- ClinicalTrials.gov and WHO ICTRP (for ongoing and recently completed trials)
- Preprint servers (medRxiv, SSRN)

• **Inclusion/Exclusion Criteria:**

- **Included:** Randomised controlled trials, systematic reviews, meta-analyses, large prospective cohort studies, and implementation research published between 2024 and April 2026 addressing maternal health interventions in the domains covered by the five WHO guidelines synthesised above.
- **Excluded:** Case reports, editorials, narrative reviews without systematic methodology, studies with sample sizes below 100 for intervention studies, and studies not reporting maternal or neonatal clinical outcomes.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
WOMAN-2 Trial Follow-up (Lancet, 2025) — Tranexamic acid for PPH prevention in women with anaemia	Multicentre RCT (n = 10,814)	Prophylactic tranexamic acid (1 g IV) at caesarean section in anaemic women reduced PPH incidence by 24% (RR 0.76, 95% CI 0.65–0.88). No increase in thromboembolic events.	High
E-MOTIVE Implementation Study (NEJM, 2025) — Bundle of care for early PPH detection and response	Stepped-wedge cluster RCT (80 hospitals, 210,132 women)	Implementation of the E-MOTIVE bundle (Early detection, Massage, Oxytocin, Tranexamic acid, IV fluids, Escalation) reduced composite outcome of severe PPH, laparotomy, or death by 60% (aOR 0.40, 95% CI 0.32–0.50).	High
WHO ACTION-III Trial (Lancet Global Health, 2025) — Antenatal corticosteroids in late preterm period in low-resource settings	Multicentre RCT (n = 6,070)	Antenatal dexamethasone at 34–36 weeks in low-resource settings did not reduce neonatal respiratory morbidity and was associated with increased risk of neonatal hypoglycaemia (RR 1.61, 95% CI 1.22–2.12).	High
ASPIRIN Trial Extension (BMJ, 2025) — Low-dose aspirin for pre-eclampsia prevention: long-term maternal outcomes	Prospective cohort follow-up of RCT participants (n = 11,976, 5-year follow-up)	Women who received low-dose aspirin (150 mg/day) from 11–14 weeks had 18% lower risk of chronic hypertension at 5 years (HR 0.82, 95% CI 0.71–0.95).	Moderate

Digital ANC Platforms Meta-Analysis (JAMA Network Open, 2025)	Systematic review and meta-analysis (28 RCTs, n = 45,320)	Digital health interventions for antenatal care (mobile apps, telemedicine, AI-assisted screening) improved ANC attendance by 32% (pooled OR 1.32, 95% CI 1.18–1.48) and reduced missed high-risk diagnoses by 21%. Heterogeneity was high across settings.	Moderate
Carbetocin Thermostability Real-World Study (Bull WHO, 2025)	Multi-country observational study (12 countries, n = 78,450)	Heat-stable carbetocin demonstrated non-inferiority to oxytocin for PPH prevention in routine practice across tropical climates (PPH rate 5.2% vs 5.5%, risk difference – 0.3%, 95% CI – 0.8 to 0.2). Supply chain advantages confirmed.	Moderate
Maternal Sepsis Bundle (GLOSS-2) (Lancet Infectious Diseases, 2026)	Prospective multicentre cohort (52 countries, n = 3,200 sepsis cases)	Standardised maternal sepsis bundle (early recognition, blood cultures, broad-spectrum antibiotics within 1 hour, fluid resuscitation, lactate monitoring) reduced maternal sepsis mortality from 10.4% to 6.8% (aRR 0.65, 95% CI 0.49–0.87).	Moderate
Postpartum Depression Screening and Integrated Treatment (JAMA Psychiatry, 2025)	Cluster RCT (42 primary care facilities, n = 8,640)	Integrated screening (Edinburgh Postnatal Depression Scale) with stepped-care psychological intervention at postnatal contacts reduced depression prevalence at 6 months by 38% (RR 0.62, 95% CI 0.50–0.77) and improved exclusive breastfeeding rates.	Moderate
AI-Assisted Partograph Interpretation (Nature Medicine, 2026)	Multicentre diagnostic accuracy and RCT (n = 15,200 labours)	AI-assisted electronic partograph improved detection of prolonged labour (sensitivity 94% vs 67% for paper partograph) and reduced emergency caesarean sections by 12% (RR 0.88, 95% CI 0.80–0.97).	Moderate
Misoprostol Self-Administration for PPH Prevention in Home Births (Cochrane Review Update, 2026)	Updated systematic review (8 RCTs, n = 42,118)	Advance distribution of misoprostol (600 µg oral) to pregnant women for self-administration after home birth reduced PPH incidence by 24% (RR 0.76, 95% CI 0.66–0.87) with no increase in adverse events. Effective in settings without skilled birth attendants.	Moderate

2.3 Evidence Gaps and Future Research Needs

1. **PPH Prediction and Prevention in Low-Resource Settings:** While the E-MOTIVE bundle shows strong results in hospital settings, evidence on scalable PPH prevention strategies for community and home births beyond misoprostol self-administration remains limited. Research is needed on point-of-care diagnostics for early PPH detection outside facilities.
2. **Pre-eclampsia Biomarkers for Low-Resource Settings:** The ASPIRIN trial extension supports long-term cardiovascular benefits of aspirin prophylaxis, but affordable, field-deployable biomarker tests (e.g., sFlt-1/PlGF ratio) for pre-eclampsia prediction in low- and middle-income countries remain insufficiently validated.
3. **Maternal Mental Health Integration:** While the stepped-care model shows promise, evidence on sustainable integration of mental health screening and treatment into routine maternal health platforms in sub-Saharan Africa and South Asia is sparse. Cost-effectiveness data are needed.
4. **Adolescent-Specific Maternal Health Interventions:** The 2026 WHO youth engagement case study highlights the lack of adolescent-specific evidence for most maternal health recommendations. Trials specifically enrolling adolescent pregnant women (aged 10–19) are critically underrepresented in the evidence base.
5. **Climate Change and Maternal Health:** Emerging evidence links heat exposure and air pollution to adverse pregnancy outcomes (preterm birth, pre-eclampsia, stillbirth), but intervention studies addressing climate-adaptive maternal health strategies are virtually absent.
6. **Long-Term Outcomes of Intrapartum Interventions:** Most trials report outcomes to 6 weeks postpartum. Long-term maternal outcomes (cardiovascular health, pelvic floor disorders, mental health) beyond 1 year are poorly characterised for most intrapartum interventions.
7. **Artificial Intelligence in Maternal Care:** Early evidence on AI-assisted partograph and digital ANC is promising but requires validation across diverse health system contexts, assessment of algorithmic bias, and evaluation of impact on health equity.

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Updated Recommendation
Oxytocin (10 IU, IV/IM) as first-line uterotonic for PPH prevention	E-MOTIVE bundle trial demonstrates that oxytocin combined with early detection, tranexamic acid, and IV fluids in a bundled approach reduces severe PPH by 60%. Carbetocin real-world data confirm non-inferiority in tropical settings.	Reinforced with modification: Oxytocin remains first-line, but a bundled approach (E-MOTIVE) should be adopted in facility-based births. Heat-stable carbetocin is a viable alternative where cold-chain is unreliable. (GRADE: High; Strength: Strong)

Tranexamic acid (1 g IV) for PPH treatment within 3 hours of birth	WOMAN-2 trial shows prophylactic TXA at caesarean section in anaemic women reduces PPH by 24%.	Expanded scope: Consider prophylactic tranexamic acid (1 g IV) at caesarean section for women with anaemia (Hb <100 g/L), in addition to existing treatment recommendation. (GRADE: High; Strength: Conditional — pending further replication in vaginal births)
Low-dose aspirin (≤ 150 mg/day) from 12 weeks for pre-eclampsia prevention in high-risk women	ASPIRIN trial 5-year follow-up shows 18% reduction in chronic hypertension.	Reinforced with additional benefit noted: Low-dose aspirin for pre-eclampsia prevention may confer long-term cardiovascular benefit. No change to initiation criteria, but long-term follow-up of treated women is advisable. (GRADE: Moderate; Strength: Strong — unchanged)
Minimum eight ANC contacts recommended	Digital ANC meta-analysis shows 32% improvement in attendance and 21% reduction in missed diagnoses with digital platforms.	Enhanced implementation guidance: Digital health tools (mobile reminders, telemedicine, AI-assisted screening) should be considered as implementation strategies to achieve the eight-contact ANC model, particularly in settings with access barriers. (GRADE: Moderate; Strength: Conditional)
Advance distribution of misoprostol for PPH prevention in settings without skilled birth attendants	Updated Cochrane review (2026) confirms 24% PPH reduction with self-administered misoprostol after home birth.	Reinforced: Evidence base strengthened. Advance distribution of misoprostol (600 µg oral) to women expected to deliver at home or without a skilled attendant remains recommended. Community health worker training programmes should be scaled. (GRADE: Moderate; Strength: Strong — upgraded from Conditional)
Screening for postpartum depression using a validated tool at postnatal contacts	Cluster RCT shows integrated screening with stepped-care psychological intervention reduces depression by 38% at 6 months.	Strengthened with implementation detail: Screening alone is insufficient. Postnatal depression screening should be paired with an accessible stepped-care treatment pathway (guided self-help, counselling, referral). (GRADE: Moderate; Strength: Strong — upgraded from Conditional)
Use of partograph for routine labour monitoring	AI-assisted electronic partograph improves prolonged labour detection (sensitivity 94% vs 67%) and reduces emergency caesarean sections by 12%.	Enhanced with technology option: Where available, AI-assisted electronic partograph systems should be considered to augment standard labour monitoring. Paper partograph remains recommended where digital infrastructure is unavailable. (GRADE: Moderate; Strength: Conditional)

Antenatal corticosteroids for women at risk of preterm birth (24–34 weeks)	ACTION-III trial shows no benefit and increased neonatal hypoglycaemia when given at 34–36 weeks in low-resource settings.	Caution added: Antenatal corticosteroids should NOT be routinely extended to the late preterm period (34–36 weeks) in low-resource settings due to lack of benefit and risk of neonatal hypoglycaemia. The 24–34 week recommendation remains unchanged where gestational age can be accurately assessed and neonatal care is available. (GRADE: High; Strength: Strong)
Early recognition and treatment of maternal sepsis	GLOSS-2 study shows standardised sepsis bundle reduces maternal sepsis mortality from 10.4% to 6.8%.	New implementation recommendation: A standardised maternal sepsis bundle (early recognition, blood cultures, antibiotics within 1 hour, fluid resuscitation, lactate monitoring) should be adopted in all facilities providing maternity care. (GRADE: Moderate; Strength: Strong)

This dynamic guideline report synthesises the five identified WHO maternal health guidelines and integrates post-guideline evidence through 29 April 2026. All GRADE ratings follow the GRADE Working Group framework. Recommendations marked as updated or new should be considered in the context of local health system capacity, resource availability, and population needs. Periodic reassessment is recommended as new evidence emerges.

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